

Connecting Raspberry Pi To External Hardware



Presented here are two simple projects to help you get started with interfacing of the GPIOs available on Raspberry Pi to external hardware. The first project demonstrates how an input data is picked up through the GPIO pin and deciphered for doing more real-world interesting jobs. In the second project, a push-to-on switch is monitored and different LEDs start blinking indefinitely corresponding to each press of the switch

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Ever since Raspberry Pi appeared in the market, this tiny credit-card-sized computer has caught the imagination of electronics and computer hobbyists around the world. The powerful Linux operating system combined with its 26 input/output (I/O) pins can do many amazing things out of the box. The best thing is that it has a general-purpose input/output (GPIO) connector through which you can connect external hardware. The basic setup and network connection were explained in EFY April and May 2013 issues, respectively. Now it's time that you explore more of Raspberry Pi with some basic experiments.

Presented here are two simple projects to help you get started with interfacing of the GPIOs available on Raspberry Pi to external hardware. The first project demonstrates how an input data is picked up through the GPIO pin and deciphered for doing more real-world interesting jobs. In the second project, a push-to-on switch is monitored, and different LEDs start blinking indefinitely corresponding to each press of the switch.

The programming language used is PHP due to its easy coding and understandability.

Project 1: Data input

In this project a switch is connected to the GPIO of Raspberry Pi and its status is continuously monitored. Once the program is executed, pressing of the



Fig. 1: Button press detected

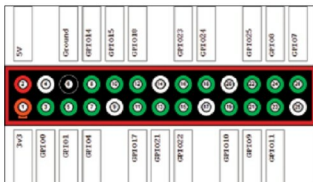


Fig. 2: Pin description of GPIO connector on Raspberry Pi

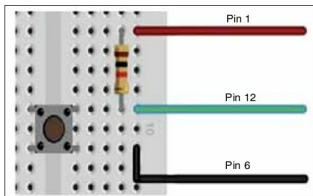


Fig. 3: Wiring diagram

button is detected as shown in Fig. 1.

Circuit and working. Fig. 2 shows the pin description of GPIO connector on Raspberry Pi. The GPIO connector has pins with different functions. The

device has seven true GPIO pins, I²C interface, SPI interface, serial TX/RX pins and PWM pins that can be used to control the external hardware.

Fig. 3 shows how to connect the GPIO of Raspberry Pi to the switch and read the input. The value of the resistor is not important—any value between 3.3 kilo-ohms and 10 kilo-ohms will work. Pin 12 monitors the status of the switch.

There are two distinct states of the GPIO pin:

1. High or 1 = 3.3 volts; provided by pin 1 through a 10-kilo-ohm resistor
2. Low or 0 = 0 volt; provided by pin 6 directly

For making pin 12 high, we use 3.3 volts (physical pin 1) available on the Raspberry board through a 10-kilo-ohm resistor connected in series. To make it low, we connect it with the ground pin (physical pin 6) of the Raspberry Pi board